

Upgrade of Hall A Møller Polarimeter

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For Hall A collaboration meeting, May 2009

Møller Polarimeter in Hall A

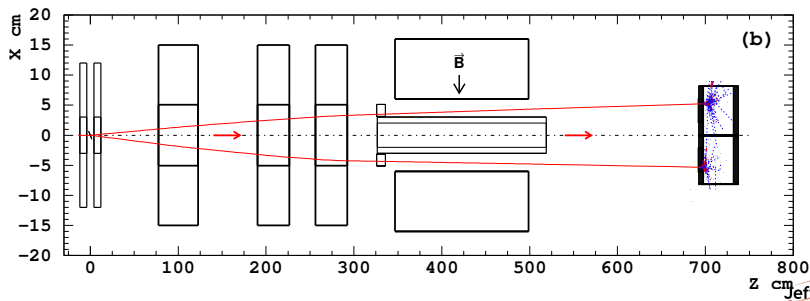
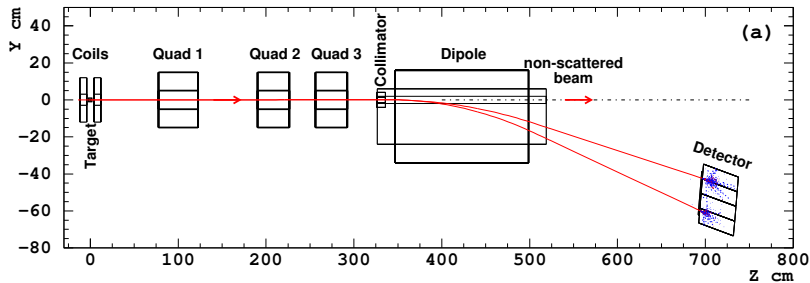
Traditional polarized electron target

- Longitudinally polarized ferromagnetic foils $7\text{-}30\ \mu\text{m}$ thick
- Relatively low magnetic field $B \sim 250\ \text{Gs}$
- Foils tilted at 20° to the beam

Spectrometer

- Quads and a dipole - minimization of $\theta - p$ correlation $\Rightarrow$ small Levchuk effect $< 1\%$
- Used at $0.6 - 6\ \text{GeV}$
- Acceptance $75^\circ < \theta_{CM} < 105^\circ$, $-10^\circ < \phi_{CM} < 10^\circ$
- Single arm rate at $0.5\ \mu\text{A} \sim 0.5\ \text{MHz}$

Møller Spectrometer



Systematic Errors

The goal for the systematic error

Variable	Error		
	OLD	Present	Upgrade goal
Target polarization	3.5%	2.0%	0.5%
Target angle	0.5%	0.5%	0.0%
Analyzing power	0.3%	0.3%	0.3%
Levchuk effect	0.2%	0.2%	0.2%
Dead time	0.3%	0.3%	0.3%
Others	-	-	0.3%
Total	3.6%	2.1%	~1.0%

Specs for the upgrade

Physics purpose, driven by PREX

- Improve the systematic error $\sim 3\% \Rightarrow \sim 1\%$
- Measurements at high beam currents $0.5\mu\text{A} \Rightarrow 50\mu\text{A}$

General Approach

- Hall C target clone (the claim $\sigma\mathcal{P}_{beam} \sim 0.5\% @ \sim 3\mu\text{A}$)
- Round target foil $\sim 1\mu\text{m}$ thick $\Rightarrow$ rate factor ~ 0.03
- Target heating $\Rightarrow \langle \mathcal{I}_{beam} \rangle < 3\mu\text{A} \Rightarrow$ duty cycle < 0.06
- Fast raster $0.7 \times 0.7\text{ mm}^2$
- Instantaneous rate $\times 2.5 \Rightarrow$ detector/electronics upgrade

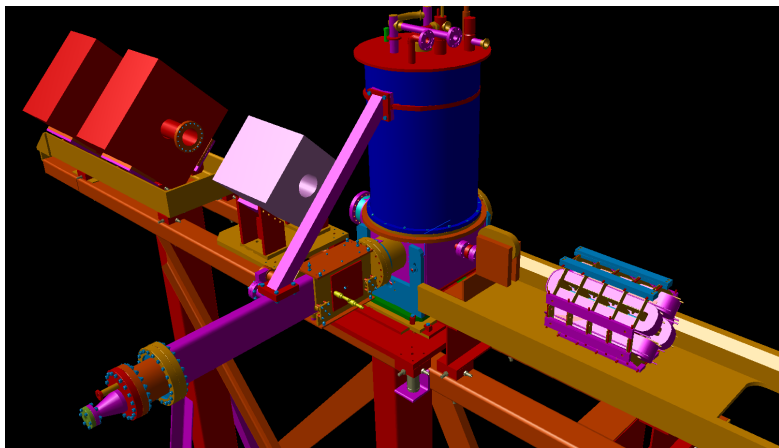
Beam Steering by 3 T Møller Coils at 1 GeV, Qds off

Magnet shifts				B	MC: Beam Deviations, mm			
δX	δY	$\delta\theta_X$	$\delta\theta_Y$		ΔX	ΔY	ΔX	ΔY
mm	mm	mrاد	mrاد	T	BPM 4A		BPM 4B	
1.0				3.0	0.50	-0.10	0.70	0.00
1.0				-3.0	0.50	0.10	0.70	0.00
	1.0			3.0	0.10	0.50	0.00	0.70
	1.0			-3.0	-0.10	0.50	0.00	0.70
		1.0		3.0	0.10	-1.40	0.20	-2.20
		1.0		-3.0	0.10	1.40	0.20	2.20
			1.0	3.0	1.40	0.10	2.20	0.20
			1.0	-3.0	-1.40	0.10	-2.20	0.20

Magnet alignment:

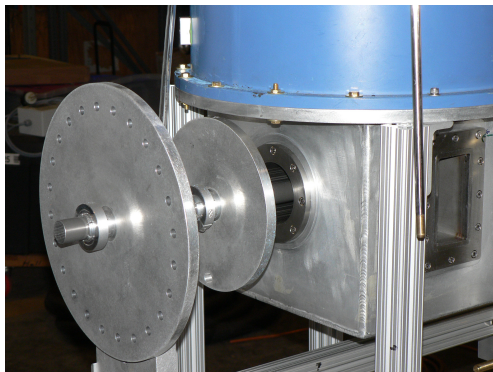
- Measure beam deflections in 4A,4B at B=3 T, B=-3 T
- Solve the equations for the magnet shifts (simple for 4B)
- Align the magnet and repeat

Target Area Design



- Special support to allow an easy magnet alignment
- Target can be aligned independently, perpendicular to the field
- 4 target positions (1, 5, 10 μm

Magnet Test, Initial Field Direction Measurement



Tests:

- Cooling, power up: by June 22
- Mapping the field: by July 1:
define the axis to ≈ 5 mrad, 3 mm

Schedule

Tests:

- Cooling, power up: by June 22
- Mapping the field: by July 1.

Installation:

- Parts expected: July 1
- Installation: July 7 - August 16
- Controls software (?)

Development:

- Electronics (FADC) parallel tests: installed in July
- Preparing for 50 μA : late fall